



EDGE AI / HPC / LLM SYSTEMS

Carmine Coppola

ML/AI Engineer & Researcher

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PORTFOLIO

carminecoppola-portfolio.vercel.app

ML/AI engineer and researcher, finishing an MSc in Machine Learning and Big Data at the University of Naples Parthenope (March 2027). My work sits between AI and systems: efficient on-device inference, HPC for scientific models and rigorous LLM evaluation, taken from prototype to something measured, tested and published. Presented at IEEE eScience and ISD.

Best Poster

IEEE eScience 2026

5 papers

and posters, 2025–26

4 venues

Belgrade, Chicago, Naples (2)

2 fellowships

FGPE 2024–25, EASY 2026

OPEN TO RESEARCH & ENGINEERING ROLES

EDGE AI

HPC

LLM EVALUATION

AGENTIC ENGINEERING

■ Work Experience

2026

(6 months)

Research Fellow – EASY Project

University of Naples Parthenope, Department of Science and Technology, Naples, Italy

Post-lauream research fellowship (DiST D.D. 85/2026) on multimodal tools for marine-environment awareness, supervised by Prof. Raffaele Montella. Started 30 May 2026, renewed for a second 3 months, until 30 November 2026. Built and validated the edge prototype end to end and took it to IEEE eScience 2026 (paper and Best Poster Award). Details under Research Projects.

2024–2025

(10 months)

Research Fellow – FGPE Project

University of Parthenope, Department of Sciences and Technologies, Naples, Italy

Fellowship within the FGPE ecosystem. Designed and implemented a component that generates gamified programming exercises with LLMs, with evaluation pipelines and dataset benchmarks. Led to the ISD 2025 paper.

2024–2026

Computer Architecture – Teaching Assistant

University of Parthenope, Department of Sciences and Technologies, Naples, Italy

Tutoring for the undergraduate Computer Architecture course: instruction-set architectures, pipelines, cache hierarchies, assembly-level programming.

■ What I Can Do

Strong

Python, PyTorch, LLM pipelines and evaluation (closed-loop generation, benchmarks, scoring), ONNX edge inference, performance instrumentation, agentic engineering with Codex and Claude Code (spec, test-gated delivery, review), scientific writing (L^AT_EX)

Working knowledge

MPI and OpenACC (collective communication, GPU directives), C/C++, scikit-learn, XGBoost, reinforcement learning (DDQN), PySpark, Raspberry Pi and V4L2 cameras, React/Next.js and TypeScript, Angular, Flutter, Docker, GitHub Actions, SLURM and HPC clusters, AWS (EC2, S3, IAM, VPC), MongoDB, MySQL

Basic

Fortran (HPC legacy refactoring), Swift, Kotlin, Java, Node.js, OpenCV, n8n automation

Tools

Jupyter, Notion, Figma, Keynote, Microsoft Office / Apple tools

Levels are self-assessed; anything not listed is not claimed.

2023–2024 **Front-End Web Developer**
(1 year) *AC Software, Lamezia Terme, Italy*

Developed responsive web applications with Angular, HTML, CSS and JavaScript in an agile team.

■ Research & Engineering Projects

Entries marked with agents (RL, API v3, app modernisation, parts of EASY) were developed with Codex and Claude Code. Architecture, validation, testing and review are mine.

EASY – Edge Node for Maritime Safety Monitoring

HPSC Lab, University of Naples Parthenope, 2025–26, open source: [EASY-Maritime-Awareness-Dashboard](#), [easy-maritime-awareness](#)

Low-cost edge platform (Raspberry Pi 4, stereo RGB, on-demand FLIR Lepton thermal, ONNX Runtime on CPU) that turns maritime observations into governed, reproducible data.

- Architected a service-oriented runtime so live and replayed sources share one code path; React dashboard and REST API on top.
- Profiled the pipeline stage by stage: persistence and request coordination, not the model, dominated latency. **3.32 s** → **0.91 s** (-72%) after instrumentation; steady state of 10 FPS per RGB view at 53% of one core and about 450 MiB; two-hour soak test with RGB at 9.55 FPS and thermal at 9.11 FPS, no restarts; **310/310** automated tests, CI.
- Found sequence-level data leakage in my own first YOLOv8n split (mAP50 0.94), documented it and rebuilt it leakage-free: mAP50 0.63 deployed, 0.68 at 960 px.

GLOwPP – GLOBO Atmospheric Model Modernisation

ISAC-CNR with University of Naples Parthenope, 2024–25 (12 months)

- Refactored legacy Fortran for modularity and portability across HPC environments.
- Replaced point-to-point MPI calls with collective routines, cutting inter-node synchronisation overhead.
- Prepared GPU readiness with OpenACC: identified hotspots and re-structured data movement.
- Built validation and regression workflows to guarantee numerical correctness. Poster at IEEE eScience 2025 (Chicago).

Certamen Artificialis Intelligentia – LLMs Solving LLM-Generated Exercises

AIED project, University of Naples Parthenope, 2024–25, open source: [Certamen-AI-Exercises](#)

- Closed-loop system in which LLMs generate programming exercises, solve and evaluate them, and iterate on difficulty and correctness; static analysis, test-case execution and scoring.
- Benchmarked ChatGPT, DeepSeek, Qwen and Gemini on Python, Java and JavaScript. Paper presented at ISD 2025 (Belgrade).

Adaptive RL for Underwater Image Enhancement

Independent project, 2026, open source: [adaptive-rl-image-enhancement](#)

Double-DQN agent (PyTorch) that learns short, interpretable sequences of image-processing actions. Official run: mean +1.55 dB PSNR on paired in-domain images, SSIM 0.83, behavioural acceptance gates passed. Out-of-domain robustness is negative and reported as unresolved.

■ Publications

An Edge Node for Citizen-Contributed Maritime Observations

INSTIL 2026 workshop, IEEE eScience 2026, Naples

Coppola, Bucciero, Perrotta, Montella. Modular Raspberry Pi 4 edge node with a node-to-observatory data-governance contract.

An Instrumented Edge Node for Dual-Sensor Maritime Safety Monitoring

Poster #312, IEEE eScience 2026, Best Poster Award (participants' selection)

Coppola, Bucciero, Perrotta, Montella. Dual-view RGB plus on-demand thermal sensing at 10 FPS on a resource-constrained board.

Certamen Artificialis Intelligentia: Evaluating AI in Solving AI-Generated Programming Exercises

ISD 2025, Belgrade, paper

Coppola et al. How ChatGPT, DeepSeek, Qwen and Gemini solve exercises generated by other LLMs.

Preserving and Improving the Legacy of eScience: The GLOBO Experience

IEEE eScience 2025, Chicago, poster

Coppola et al. HPC refactoring, MPI optimisation and GPU readiness of the GLOBO model.

Directed Acyclic Graph on Cross-Application Programmable I/O

CINI HPC Summer School, Naples 2025, poster (co-author)

Perrotta et al. Streaming flavour for scientific workflows.

app@uniparthenope – Official University Mobile Application

Institutional publication, 2024

Coppola et al. Architecture and deployment of the official Android and iOS app.

UniParthenope API v3 Gateway

University of Naples Parthenope, 2026, private repository

Backend gateway built from scratch that keeps all 91 operations of the legacy API unchanged, so the existing app does not notice the refactoring. Developed with Codex and Claude Code; pre-deploy security audit, hardening and deployment on the cluster; automated test suite.

app@uniparthenope – Official University App

University of Naples Parthenope, 2024–26

Flutter app for Android and iOS (careers, courses, fees, weather, biometric login), published as an official institutional release. In 2026 I modernised the whole codebase with Codex and Claude Code, backed by unit tests for auth and career state.

Hi-WeFAI – AI Hydrometeorological Workflow Tutorial

University of Naples Parthenope, 2025, open source: [Hi-WeFAI_tutorial](#)

Operational tutorial: radar ingestion, transformer-based precipitation nowcasting, hydrological coupling with PERFECT-M, and reproducible Docker/HPC deployment guides.

Other work

open source on [GitHub](#)

- Diachronic semantic analysis of Italian: CBOW embeddings per decade (Google Books n-grams), Procrustes alignment, SLURM training.
- PySpark SMS spam classifier benchmarked on a Hadoop cluster.
- OpenUsage for VS Code: extension showing spend, limits and usage of AI coding tools, with CI packaging.

■ Conferences & Dissemination

2026	IEEE eScience 2026, Best Poster Award <i>Naples, 28 Sep–2 Oct, Poster #312, awarded 1 Oct</i> INSTIL workshop paper and poster on the EASY edge node. Best Poster Award (participants' selection), signed by the General Chair.
2025	ISD 2025, Information Systems Development <i>Belgrade, 3–5 September, Paper</i> LLMs solving AI-generated programming exercises.
2025	IEEE eScience 2025 <i>Chicago, USA, Poster</i> GLOWPP: modernisation of the GLOBO model (MPI, OpenACC).
2025	CINI Summer School on HPC <i>Naples, 16–20 June, Poster</i> Co-author, “Directed Acyclic Graph on Cross-Application Programmable I/O”.
2024–2025	Seminar: AI & ML with Python <i>University of Naples Parthenope, Instructor</i> Full ML workflow with scikit-learn and PyTorch on real datasets.

■ Education

MSc, Machine Learning and Big Data

University of Naples Parthenope • 2024–present
Expected March 2027. Statistical learning, deep neural networks, big data infrastructures, AI system design.

BSc, Computer Science

University of Naples Parthenope • 2020–2024
Graduated 25 June 2024, **97/110**.

Technical Diploma, Electrotechnics

I.T.I.S Eugenio Barsanti • 2015–2020
Electronics, electrical systems, industrial automation.

■ Certifications

English B2

Language certificate
Used daily for papers, posters and talks.

AWS Academy Cloud Foundations

AWS Academy • 2024–25
EC2, S3, IAM, VPC and scalable architecture design. Also AWS Educator Accreditation (2024).

Kaggle Learn

Python, Intro to ML, Intermediate ML, ML Explainability • 2023–26
Pipelines, cross-validation, XGBoost, data-leakage prevention, SHAP and permutation importance.

iOS Development Basic Course

University of Naples Parthenope • 2022
Swift, Xcode, UIKit fundamentals, MVC.

■ Languages

Italian

Native

English

B2 (certified); papers, posters and talks in English

■ Personal Data Processing

According to the artt. 46 and 47 of the DPR 445/2000 (Italian Law) and aware that false statements are punishable according to the penal code and by the special laws on this subject. In compliance with the GDPR and Italian Legislative Decree no. 196 dated 30/06/2003, I hereby authorize the recipient of this document to use and process my personal details for the purpose of recruiting and selecting staff and I confirm to be informed of my rights in accordance to art. 7 of the above mentioned Decree.